Sellekni · Business Model Canvas

SolveX 3.0 · Challenge 1 · Road emergencies in Algeria · A phone-based crash alert that reaches the emergency operations room by internet or plain SMS, with an estimated risk and a hospital suggestion.

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The model in one sentence. The apps are free for everyone who sends an alert and, during the pilot, for the emergency services that receive it. Money comes from paid hardware (an OBD dongle and the Sellekni Box) for drivers and fleets who want stronger crash detection, then from institutional licences and insurers, with non-dilutive funding to start. Payers and prices are still to be validated.

8 · KEY PARTNERS

Who could help us?

- **Civil Protection** (14 / 1021) ops rooms: co-design, ground truth
- **Health ministry, SAMU, hospital EDs:** capability data, clinical review
- **Telecom operators:** SMS short code, coverage maps, LTE SIMs for the box
- **Emergency physicians, universities:** score calibration
- **Electronics makers and assemblers:** box prototype and production
- **ARPCE:** equipment homologation
- **DNSR:** statistics, road-safety campaigns
- **Incubators, startup-label committee**
- Later: **insurers, fleets** for distribution
- TO APPROACH · NONE AGREED

7 · KEY ACTIVITIES

What must we do?

- Build the app, gateway and dashboard
- Record real drives, tune the detector
- Verify hospitals; calibrate the score with physicians
- **Hardware:** box prototype, C encoder, homologation
- Operator training; legal (law 18-07 / 25-11)

6 · KEY RESOURCES

What do we need?

- **Built:** SLK1 protocol and rules engine (67 tests + Python cross-check), crash detector with replays (21 tests), dashboard, witness and hospital views in AR / FR / EN VERIFIED
- **Designed:** Sellekni Box (ESP32), OBD link
- **To build:** verified hospital registry, hosting in Algeria, real driving data

2 · VALUE PROPOSITION

What problem do we solve?

- **Victim:** if you can't call, your phone alerts with your exact position. 30 s to say "I'm OK", then it sends itself. Works by plain SMS when data is weak; queues when there is no signal.
- **Witness:** report in 5 taps in the app, or 4 questions from a web link; no account.
- **14 operator:** one merged, located report with an estimated risk and its reasons; hospitals ranked by capability and travel time. The human decides; every action is logged.
- **Hospital ED:** a pre-alert before the patient arrives, acknowledged.
- **Hardware buyer:** the OBD dongle sharpens the speed drop; the Box's ±200 g sensor sees the real crash pulse that phones clip, with its own GNSS, LTE and backup power.
- **We don't** replace 14 / 1021, diagnose, or claim real-time bed data.

4 · CUSTOMER RELATIONSHIPS

How do we support them?

- Weekly co-design and a training day with operators; a named contact per partner
- Quiet app: no ads, no tracking outside an alert; SMS receipt to the sender
- "False alarm" feedback loop tunes the detector
- **Hardware:** install guide, firmware updates, warranty H

3 · CHANNELS

How do we reach them?

- Pilot: APK, witness link, dashboard in a browser
- Later: Play Store (SMS-permission policy or composer fallback)
- Civil Protection days, driving schools, DNSR campaigns
- **Hardware:** fleet onboarding, insurer packs, car-accessory shops, online H

1 · CUSTOMER SEGMENTS

Who are they?

- **Free users:** drivers and passengers with Android phones (≈ 87 % of phones in Algeria) STATCOUNTER ; witnesses
- **Free receivers (pilot):** Civil Protection 14 / 1021, Gendarmerie 1055, Police 17, SAMU where present; hospital EDs
- **Hardware buyers:** drivers who want stronger detection; fleets and transport operators (buses, coaches, lorries) H
- **Institutional payers:** the State (B2G); motor insurers, ≈ 41.6 % of 2024 premiums TRÉSOR VIA ATLAS MAG.
- **Pilot early adopters:** one wilaya's ops room, one university hospital, one fleet

9 · COST STRUCTURE

What will it cost?

- **Fixed:** team time (the main cost); hosting in Algeria; map and routing hosting
- **Per incident:** about one SMS receipt and one hospital pre-alert, plus gateway SIMs
- **Hardware:** parts (ESP32, ±200 g sensor, LTE/GNSS modem, power), enclosure, box SIM data, homologation (Executive Decree 26-97), warranty
- **One-off:** medical and legal review, trademark (INAPI), test phones
- No supplier quotes yet, so no figures H

5 · REVENUE STREAMS

How will it be funded?

- **Apps: always free for citizens.** Nobody pays to be alerted.
- PAID **OBD dongle:** entry-level hardware sale to drivers and fleets
- PAID **Sellekni Box:** premium hardware sale, with an optional connectivity subscription for its SIM H
- **B2G:** licence and support contract for the ops-room software H
- **Fleets and insurers:** crash notification and data export subscription H
- **Now:** prizes, incubators, Algeria Startup Fund after labelling ROUTE VERIFIED
- No price set before interviewing 3 fleet managers and 2 insurers.

HARDWARE PLAN · ROADMAP · PAID TIER

Free app, paid hardware, one packet

FREE · TODAY
Phone app
accelerometer, gyro, GPS; saturates on hard impacts

FREE · TODAY
Witness
in the app or a web link

PAID · ROADMAP
OBD dongle
wheel speed from the diagnostic port

PAID · ROADMAP
Sellekni Box
±200 g sensor, own GNSS, LTE, backup power

all sources → **same 160-char packet** → **same server** → a human decides

SCHEMA · SELLEKNI BOX (ESP32)

Reference design and firmware flow

__BOX__

Acceptance test: the C encoder reproduces our 302 valid and 134 invalid test packets byte for byte. LTE because Algeria plans to retire 2G/3G by 2029.

SCHEMA · OBD LINK

What the car's diagnostic port adds

Car OBD-II port pin 16: +12 V → **BLE dongle** ELM327-type → **Phone app** reads PID 0D → **Same detector** sharper speed drop

Can:
- vehicle speed, PID 0D (km/h)
- engine speed, PID 0C
- faster speed than 1 Hz GPS

Can't:
- read "airbag deployed": no standard signal
- B0001 is a circuit fault, not a deployment

Status: box and OBD link are designed, not built, not tested in a vehicle, not homologated. No hardware price is set.

Sources: DNSR 2025 via Atlas Magazine; Direction Générale du Trésor via Atlas Magazine (insurance 2024); StatCounter (Android share); decrees 20-254 / 25-311 (startup label); Executive Decree 26-97 (homologation).